Applied Statistics (MATH10096)

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- Most of the models covered so far have been for situations in which the predictor variables are continuous variables but there are many situations in which the predictor variables are more qualitative in nature, and serve to divide the responses into groups.
- These variables are known as factor variables or simply factors.
- The different categories of the factor are known as levels of the factor.
- For example, levels of the factor 'eye colour' might be 'blue', 'brown', 'grey' and 'green', so that we would refer to eye colour as a factor with four levels.

Practical modelling with factors and interactions

General context

- Factor variables are handled using dummy (or indicator) variables.
- Each factor variable can be replaced by as many dummy variables as there are levels of the factor — one for each level of the factor.
- For each response observation, only one of these dummy variables will be non-zero: the dummy variable for the single level that applies to that response.
- Consider an example to see how this works in practice: nine laboratory rats are overfed, dividing them into three groups of three: ‘fat’, ‘very fat’ and ‘enormous’.
- Their blood insulin levels are then measured ten minutes after being fed a standard amount of sugar.
- The investigators are interested in the relationship between insulin levels and the factor ‘rat size’.

Practical modelling with factors and interactions

General context

- A model could be set up in which the predictor variable is the factor 'rat size', with the three levels 'fat', 'very fat' and 'enormous'.

- Writing y_i for the i^{th} insulin level measurement, a suitable model might be:

$$\mathrm{E}(Y_i) \equiv \mu_i = \begin{cases} \beta_0 & \text{if rat } i \text{ is fat,} \\ \beta_1 & \text{if rat } i \text{ is very fat,} \\ \beta_2 & \text{if rat } i \text{ is enormous.} \end{cases}$$

- This is easily written in linear model form, using a dummy predictor variable for each level of the factor:

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

Practical modelling with factors and interactions

Identifiability

- When modelling with factor variables, model identifiability becomes an important issue.
- It is quite easy to set up models, involving factors, in which it is impossible to estimate the parameters uniquely, because an infinite set of alternative parameter vectors would give rise to exactly the same expected value vector.
- To illustrate this, consider again the fat rat example, but suppose that we wanted to formulate the model in terms of an overall mean insulin level, α, and deviations from that level, β_j, associated with each level of the factor.
- The model would be something like:

$$\mu_i = \alpha + \beta_j, \quad \text{if rat } i \text{ is rat size level } j, \qquad j \in \{0, 1, 2\}.$$

- Here j is 0, 1 or 2, corresponding to 'fat', 'very fat' or 'enormous'

- The problem with this model is that there is not a one-to-one correspondence between the parameters and the fitted values, so that the parameters can not be uniquely estimated from the data.
- Consider any particular set of α and $\boldsymbol{\beta}$ values, giving rise to a particular $\boldsymbol{\mu}$ value: any constant c could be added to α and simultaneously subtracted from each element of $\boldsymbol{\beta}$ without changing the value of $\boldsymbol{\mu}$.
- For instance, $\mu_1 = \alpha + \beta_0 = (\alpha + c) + (\beta_0 - c)$ and the same for $\mu_2, \ldots, \mu_9$.
- Hence there is an infinite set of parameters giving rise to each $\boldsymbol{\mu}$ value, and therefore the parameters can not be estimated uniquely.
- The model is not identifiable!

- This situation can be diagnosed directly from the model design matrix

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

- The columns of the model matrix are not linearly independent: the first column is the sum of the second, third, and fourth columns.

- This lack of full column rank means that the formulae for finding the least squares parameter estimates break down.

- One possible solution to the identifiability problem is to set one of the unidentifiable parameters to zero, which requires only that the model is re-written, without the zeroed parameter.

- For example, in the fat rat case, we could set α to zero, and recover the original identifiable model we started with.

- When specifying models with factor variables in R, the software automatically imposes an identifiability constraint.

- By default, and contrary to what we just mentioned, the constraint is that the parameter for the 'first' level of the factor is set to zero.

Practical modelling with factors and interactions
Identifiability

- Let us use dummy variables and write down the same model.
- Let $x_{1i} = 1$ if rat i is very fat and $x_{1i} = 0$ otherwise and, similarly, $x_{2i} = 1$ if rat i is enormous and $x_{2i} = 0$ otherwise.
- Then,

$$\mu_i = \alpha + \beta_1^* x_{1i} + \beta_2^* x_{2i}.$$

- For fat rats, since $x_{1i} = x_{2i} = 0$, we have that $\mu_i = \alpha$.
- For very fat rats, $x_{1i} = 1$ and $x_{2i} = 0$, so $\mu_i = \alpha + \beta_1^*$.
- For enormous rats, $x_{1i} = 0$ and $x_{2i} = 1$ and hence $\mu_i = \alpha + \beta_2^*$.

Practical modelling with factors and interactions

Identifiability

- Suppose that rat i is fat and rat i' is very fat, then

$$\mu_{i'} - \mu_i = \alpha + \beta_1^* - \alpha = \beta_1^*,$$

  and so this parameter represents the expected difference in insulin levels between a very fat and a fat rat.

- Similarly, β_2^* represents the expected difference in insulin levels between an enormous rat and a fat rat.

- Further note that if rat i' is very fat and rat i'' is enormous then

$$\mu_{i'} - \mu_{i''} = (\alpha + \beta_1^*) - (\alpha + \beta_2^*) = \beta_1^* - \beta_2^*,$$

  i.e., the expected difference in insulin levels for a very fat rat and an enormous rat is $\beta_1^* - \beta_2^*$.

Practical modelling with factors and interactions

Multiple factors

- Continuing the fat rat example, suppose that now two factor variables are involved: it might be that, beyond the size of the rat, the sex of the rats is also a factor in insulin production.
- The appropriate model is:

$$\mu_i = \alpha + \beta_j + \gamma_k \text{ if rat } i \text{ is rat size level } j \text{ and sex } k, \quad j \in \{0,1,2\}, \quad k \in \{0,1\}.$$

- Here k is 0 or 1 for male or female.
- Written out in full (assuming the rats are MMFFFMFMM) the model is

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_0 \\ \beta_1 \\ \beta_2 \\ \gamma_0 \\ \gamma_1 \end{pmatrix}.$$

Practical modelling with factors and interactions

Multiple factors

- It is immediately obvious that the model matrix is of column rank 4, implying that two constraints are required to make the model identifiable: column 5 is column 1 minus column 6, while column 2 is column 1 minus columns 3 and 4.

- An obvious pair of constraints is to set $\beta_0 = \gamma_0 = 0$ (and this aligns with what R does: exclude the first level of each factor variable), so that the full model is

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_1 \\ \beta_2 \\ \gamma_1 \end{pmatrix}.$$

- Note that in this case the intercept α represents the expected insulin levels for a fat male rat.

Practical modelling with factors and interactions

'Interactions' of factors

- In the examples considered so far, the effect of factor variables has been purely additive.
- It is, however, possible that a response variable may react differently to the combination of two factors, than would be predicted by simply adding the effect of the two factors separately.
- For example, if examining patient blood cholesterol levels, we might consider the factors 'hypertensive' (yes/no) and 'diabetic' (yes/no).
- Being hypertensive or diabetic would be expected to raise cholesterol levels, but being both is likely to raise cholesterol levels much more than would be predicted from just adding up the apparent effects when only one condition is present.
- When the effect of two factor variables together differs from the sum of their separate effects, then they are said to interact, and an adequate model in such situations requires interaction terms.
- Put another way, if the effects of one factor change in response to another factor, then the factors are said to interact.

Practical modelling with factors and interactions

'Interactions' of factors

- Let us continue with the fat rat example, but now suppose that how insulin level depends on size varies with sex.
- An appropriate model is then

$$\mu_i = \alpha + \beta_j + \gamma_k + \delta_{jk} \text{ if rat } i \text{ is rat size level } j \text{ and sex } k,$$

- Here δ_{jk} terms are the parameters for the interaction of rat size and sex.
- Writing this model out in full it is clear that it is spectacularly unidentifiable:

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_0 \\ \beta_1 \\ \beta_2 \\ \gamma_0 \\ \gamma_1 \\ \delta_{00} \\ \delta_{01} \\ \delta_{10} \\ \delta_{11} \\ \delta_{20} \\ \delta_{21} \end{pmatrix}.$$

Practical modelling with factors and interactions
'Interactions' of factors

- For this simple example, with rather few rats, we now have more parameters than data.
- One possibility (the default in R) to identify the model is to set

$$\beta_0 = \gamma_0 = \delta_{00} = \delta_{01} = \delta_{10} = \delta_{20} = 0,$$

resulting in

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_1 \\ \beta_2 \\ \gamma_1 \\ \delta_{11} \\ \delta_{21} \end{pmatrix}.$$

Practical modelling with factors and interactions

Factor continuous interactions

- Suppose we had access to the exact weight, w_i, of the rats, rather than only its classification into one of the three groups previously defined.
- In this case an appropriate model might be

$$\mu_i = \alpha + \gamma_j + \beta w_i + \delta_j w_i \text{ if rat } i \text{ is sex } j.$$

- That is to say insulin levels vary linearly with weight, but in a different way for male and female rats.
- Here γ_j would be the 'main effect' of sex, βw_i the 'main effect' of weight, and $\delta_j w_i$ is the weight-sex 'interaction'.
- Recalling the assumed ordering of rats, MMFFFMFMM, then an identifiable version of the model is

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & w_1 & 0 \\ 1 & 0 & w_2 & 0 \\ 1 & 1 & w_3 & w_3 \\ 1 & 1 & w_4 & w_4 \\ 1 & 1 & w_5 & w_5 \\ 1 & 0 & w_6 & 0 \\ 1 & 1 & w_7 & w_7 \\ 1 & 0 & w_8 & 0 \\ 1 & 0 & w_9 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \gamma_1 \\ \beta \\ \delta_1 \end{pmatrix}.$$

Practical modelling with factors and interactions

Factor continuous interactions

- It is very easy to work with factor variables in R: all that is required is that you let R know that a particular variable is a factor variable.
- For example, suppose z is a variable declared as follows:

```
z <- c(1, 1, 1, 2, 2, 1, 3, 3, 3, 3, 4)
z
##  [1] 1 1 1 2 2 1 3 3 3 3 4
```

- And it is to be treated as a 4 level facton: the function `as.factor()` will ensure that `z` is treated as a factor:

```
z <- as.factor(z)
z
##  [1] 1 1 1 2 2 1 3 3 3 3 4
## Levels: 1 2 3 4
```

- When a factor variable is printed, a list of its levels is also printed: this provides an easy way to tell if a variable is a factor variable.
- Note also that the digits of `z` are treated purely as labels: the numbers 1 to 4 could have been any labels.

Practical modelling with factors and interactions

Factor continuous interactions

- For example, `x` could be a factor with 3 levels, declared as follows:

```
x <- c("A","A","C","C","C","er","er")
x

## [1] "A"  "A"  "C"  "C"  "C"  "er" "er"

x <- as.factor(x)
#check reference level (the row that only contains zeroes)
contrasts(x)

##    C er
## A  0  0
## C  1  0
## er 0  1
```

- Suppose we want to change the reference level from A to er. We can do that through the use the `relevel` function as follows:

```
x <- relevel(x, ref = "er")
x

## [1] A  A  C  C  C  er er
## Levels: er A C

contrasts(x)

##    A C
## er 0 0
## A  1 0
## C  0 1
```